

NEXUS

The Floating Desktop AI Assistant

A cross-platform, Siri-like floating overlay assistant that lives on your desktop.

The Problem

- Modern AI assistants are either **trapped in a browser tab** or require **heavy local resources**.
- They lack seamless, non-intrusive integration with the user's desktop workflow.
- **Privacy concerns:** Cloud-based audio processing sends your voice to third-party servers.

The Solution: NEXUS

- **Thin Client:** Tauri v2 + Rust + React/TS. (Idle RAM < 90 MB).
- **Fat Server:** n8n supervisor + Ollama for intelligent intent routing and LLM processing.
- **Privacy First:** Wake-word and Speech-to-Text run **locally**. Only text crosses the network.

Key Highlights

- **Transparent & Frameless:** Always-on-top overlay with *region-aware click-through* (clicks fall through transparent pixels).
- **Low-Power Wake Word:** openWakeWord in pure Rust (~0.5% CPU, fully offline).
- **Streaming TTS:** Gapless WebAudio playback for instant responses.
- **Zero Idle CPU:** No background JS loops; entirely event-driven.

Architecture Topology

- **Thin Client (Rust + React):**
 - openWakeWord KWS → Wake Event
 - Local STT (faster-whisper) → WSS Bridge
- **Fat Server (GPU Host):**
 - Webhook → n8n Master Supervisor
 - Intent Router → Ollama (LLM) → TTS (Piper) → WSS

Intelligent Intent Router (n8n)

- **Sub-canvas Workflows:**

- `email.summarize` : Summarizes your recent emails (IMAP/Graph API).
- `github.pr_check` : Reviews PR status & risk (GitHub REST).
- `calendar.peek` : Narrates your next 3 events.
- `general.chat` : Fast, free-form conversational replies.

Built For Every Platform

- **Windows 10/11:** NSIS installer; WebView2 bootstrapper.
- **macOS (Silicon & Intel):** Universal `.dmg`, notarized; `LSUIElement` hides dock.
- **Linux (X11 & Wayland):** AppImage & `.deb`; requires compositor for true transparency.

Why NEXUS Wins

- **Performance:** Native Rust backend ensures a minimal resource footprint.
- **Privacy:** Your voice never leaves your machine. Text-only network protocol.
- **Extensibility:** n8n blueprints allow infinite server-side capabilities without client updates.
- **UX:** Region-aware click-through makes it feel like magic.

Thank You

NEXUS — The Future of Desktop AI